AI Agents intensive

Школа анализа данных

Zaytseva Alena,
AI Lead at Yandex Lavka

About the intensive

Team

- Yandex Lavka
- Alice & Smart Devices
- Search Portal
- Yandex Cloud

Goal

- Learn to quickly build an MVP/personal assistant
- Learn to turn a prototype into a production-ready system

What an Agent Actually Does

01

Day 1

- Intro to AI Agents
- LLMs
- Agent Tools & MCP

02

Day 2

- Memory
- Guardrails

03

Day 3

- AI Agent Workflows
- Multi-Agent & Multi-Modal Systems

04

Day 4

- AI Agent Evaluation

05

Day 5

- Production Engineering of AI Agents

Test

Школа анализа данных

Lecture 1.1

Intro to AI Agents LLM

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Intro to AI agents

01 AI Agent Applications

02 The Evolution of Human–LLM Interaction

03 What Is an AI Agent

LLMs

01 What Is an LLM

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03 How LLMs are Trained

04 What LLMs Can and Cannot Do

05 About over-engineering

Practice

Using LLMs, Building a Simple Agent

AI Agent Applications

Knowledge & Productivity

- Personal assistants (research, decision support)
- Alice AI, ChatGPT, Perplexity,

Action & Execution

- Browser agents (navigate, click, fill forms)
- Developer agents (Cursor, GitHub Copilot)

Business & Interaction

- Customer support
- Operations automation

Interactive Worlds

- Game NPCs with adaptive behavior and dialogue

The Evolution of Human–LLM Interaction

Interaction	Description	Pattern example	
Prompting	LLM output has no impact on program flow	process_llm_output(llm_response)	
Router	LLM determines the basic control flow	if llm_decision(): path_a() else: path_b()	
Tool Calling	LLM selects the function and arguments	run_function(llm_chosen_tool, llm_chosen_args)	
Multi-step Agent (Autonomous Agent)	LLM output controls iterations and continuation of program execution	while llm_should_continue(): execute_next_step()	
Multi-Agent System	One agent process can trigger another agent process	if llm_trigger(): execute_agent()	

What Is an AI Agent?

OpenAI Agents SDK presentation [1]:

An agent is an AI application consisting of

- a model equipped with
- instructions that guide its behavior,
- access to tools that extend its capabilities,
- encapsulated in a runtime with a dynamic lifecycle.

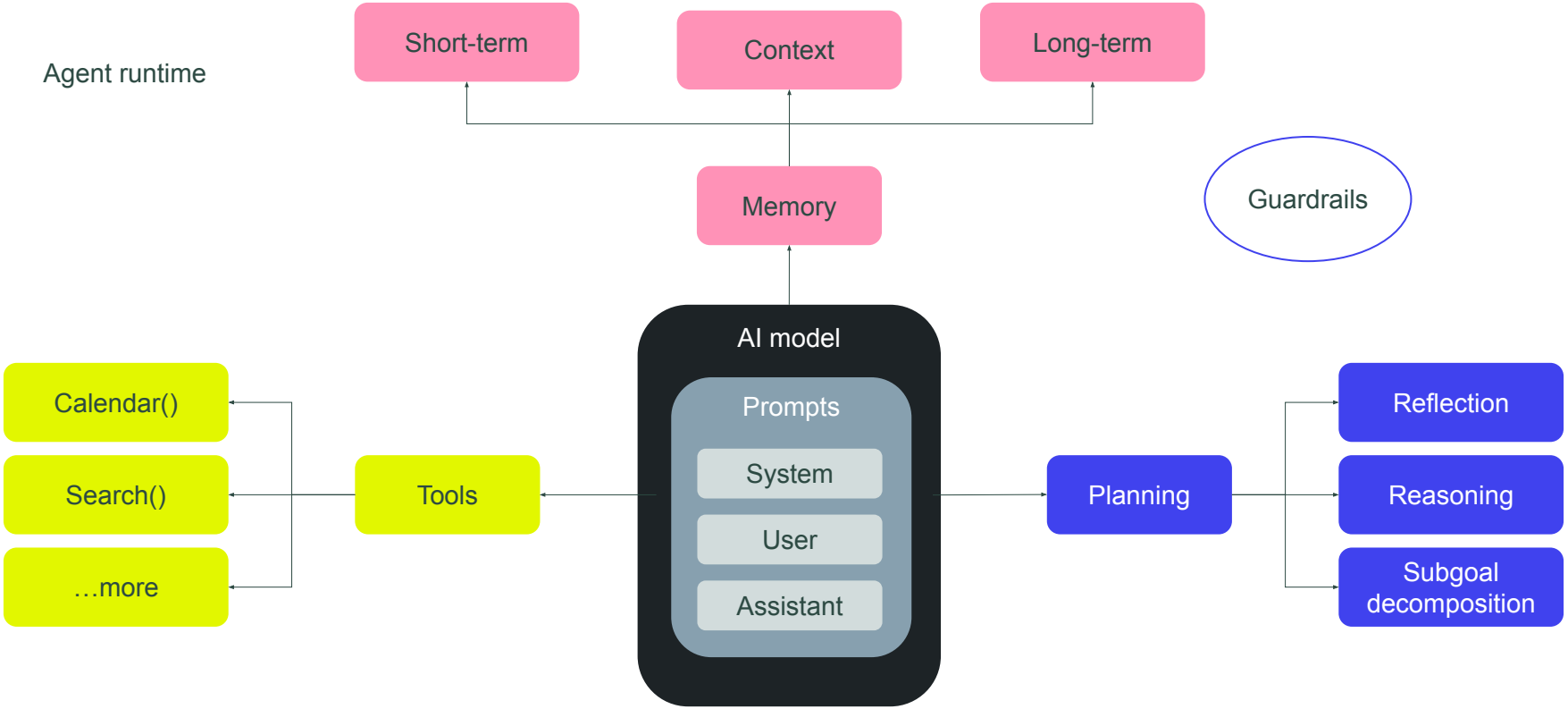
Lilian Weng, LLM Powered Autonomous Agents [2]:

Agent = LLM + memory + planning skills + tool use

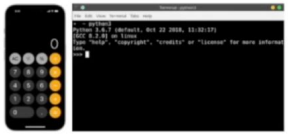
[1] https://www.youtube.com/watch?v=joHR2pmxDQE
[2] https://lilianweng.github.io/posts/2023-06-23-agent/

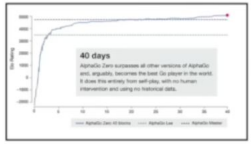
What Is an AI Agent?

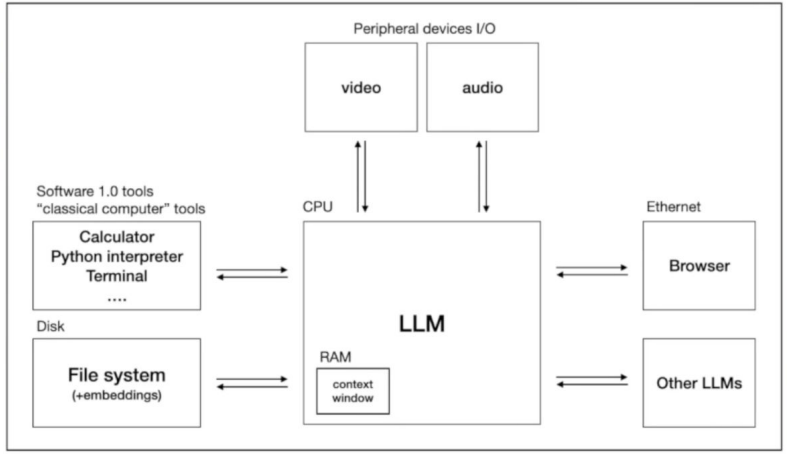
Agent = runtime(AI model + prompts + tools + memory + guardrails + planning skills)

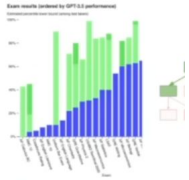


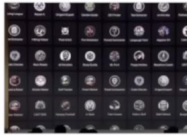
LLM as an Operating System

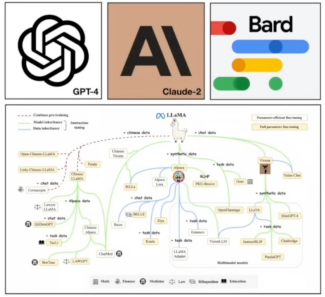






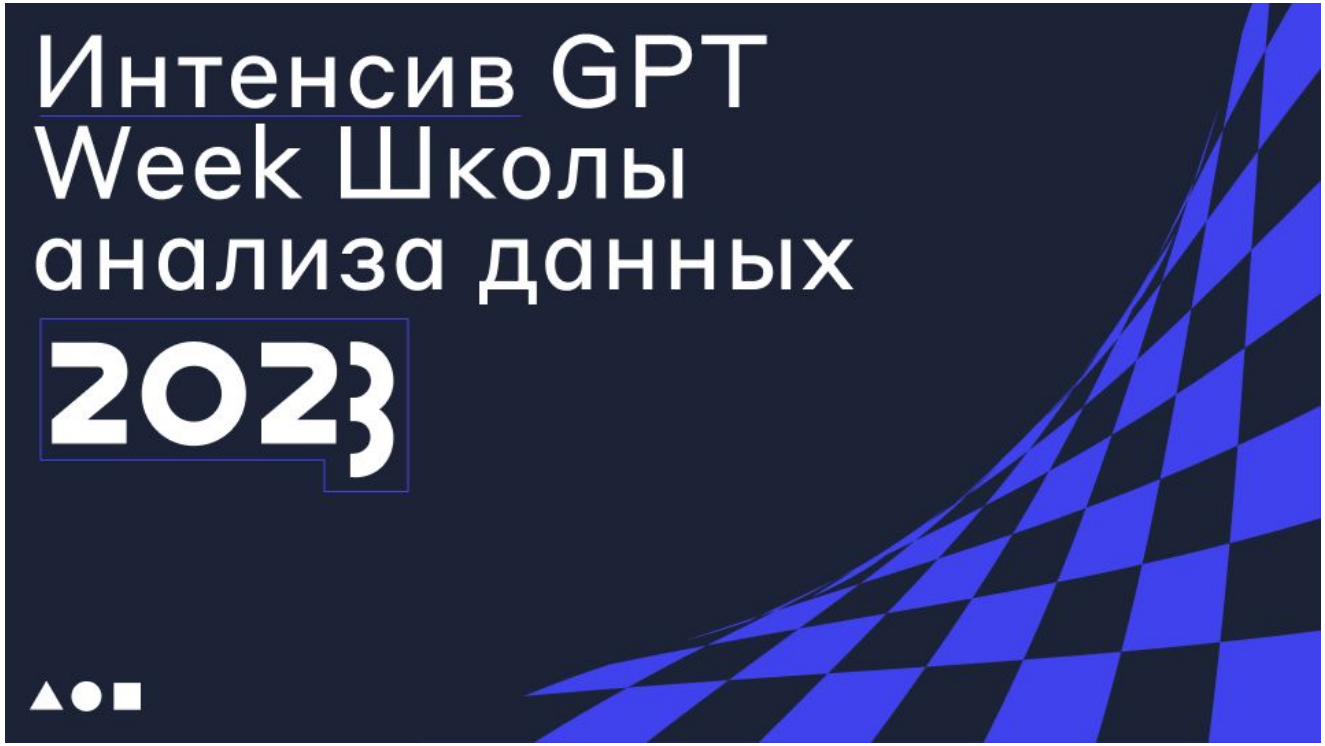






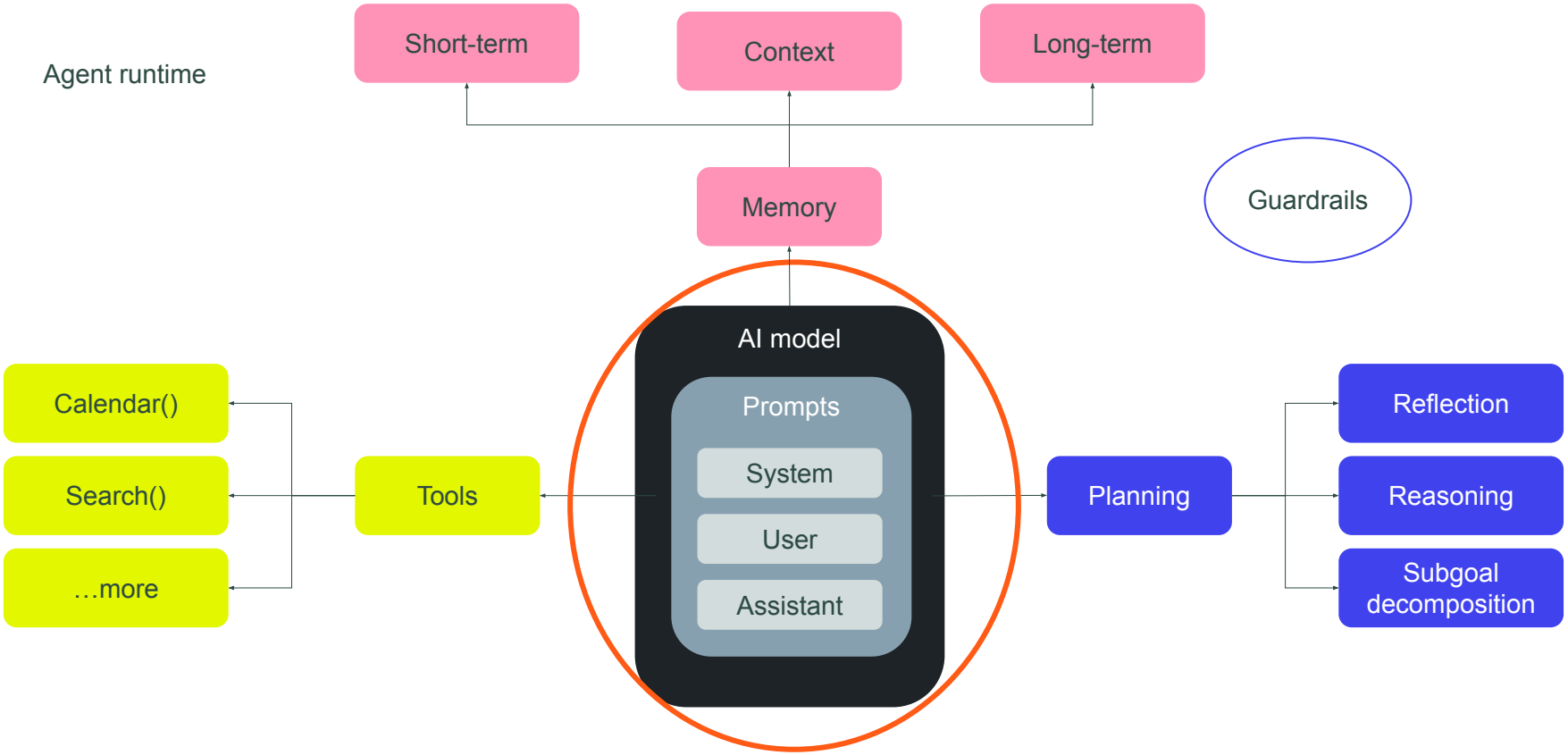
Andrej Karpathy, Intro to Large Language Models
https://www.youtube.com/watch?v=zjkBMFhNj_g

LLMs



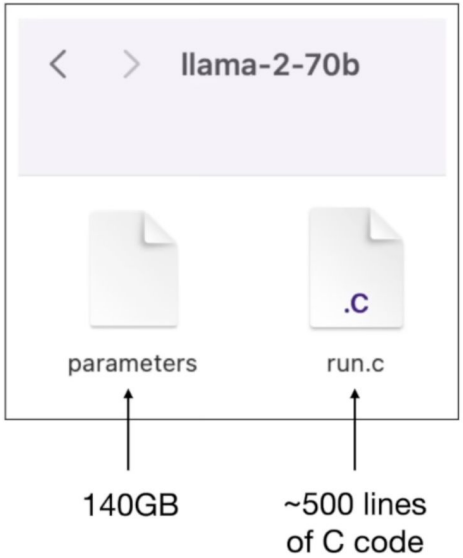
GPT Week https://shad.yandex.ru/gptweek

LLM — The Brain of the Agent



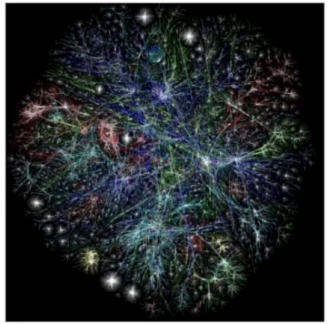
What Is an LLM

Large Language Model (LLM) is a neural network model trained on large corpora of text data that predicts the next token based on the context

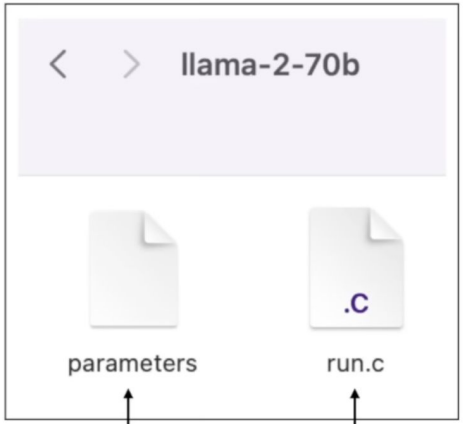


What is an LLM: Weights

Large Language Model (LLM) is a neural network model trained on large corpora of text data that predicts the next token based on the context

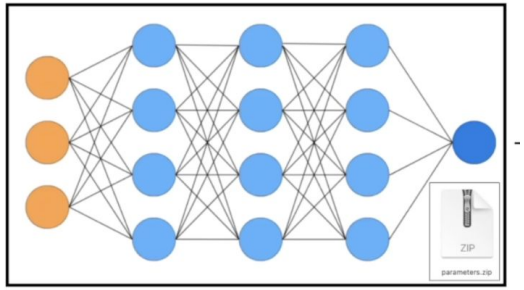


Chunk of the internet, ~10TB of text



140GB

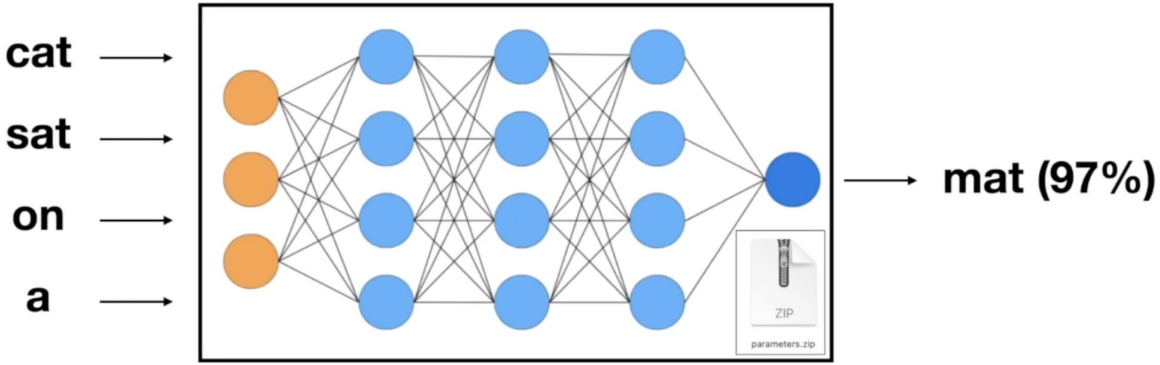
~500 lines of C code



What is an LLM: Next Token Prediction

Large Language Model (LLM) is a neural network model trained on large corpora of text data that predicts the next token based on the context

Start with words



Tokens: Where Do They Come From

How to encode “I was reading an interesting book in New York”

Character encoding

- I=1, _=2, w=3, a=4, s=5, r=6, e=7, d=8, i=9, n=10, g=11, t=12, b=13, o=14, k=15, N=16, Y=17
- **1 2 3 4 5 2 6 7 4 8 9 10 11 2 4 10 2 9 10 12 7 6 7 5 12 9 10 11 2 13 14 14 15 2 9 10 2 16 7 3 2 17 14 6 15**

Word encoding

- I=1, was=2, reading=3, an=4, interesting=5, book=6, in=7, New=8, York=9
- **1 2 3 4 5 6 7 8 9**

Tokenization

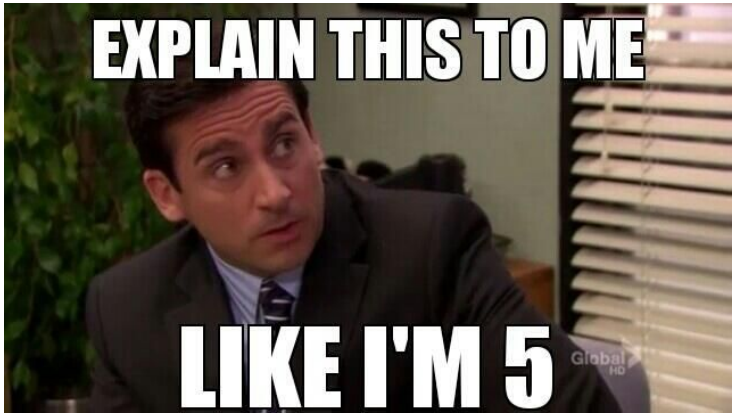
- I=1, was=2, read=3, ing=4, an=5, interest=6, book=7, in=8, New_York=9
- **1 2 3 4 5 6 4 7 8 9**

Tokens: Special Tokens

- Reserved tokens with predefined functions
- Do not represent normal text
- Control model behavior and structure inputs/outputs

Examples

- EOS tokens: <eos>, <|endoftext|> — stop generation
- Chat roles: <|user|>, <|assistant|>, <|system|>
- Control tokens: <translate>, <summarize>, <formal>



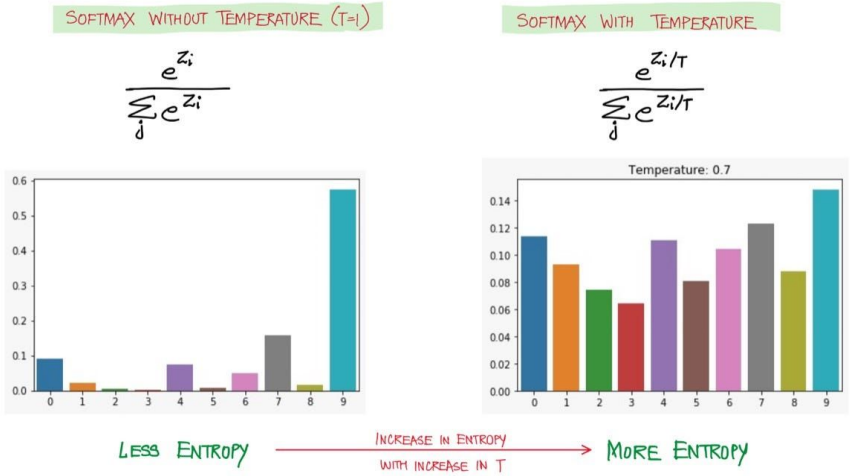
What is an LLM: Next Token Prediction

Large Language Model (LLM) is a neural network model trained on large corpora of text data that predicts the next token based on the context

- autoregression
- context window length

Predicting Token Sequences

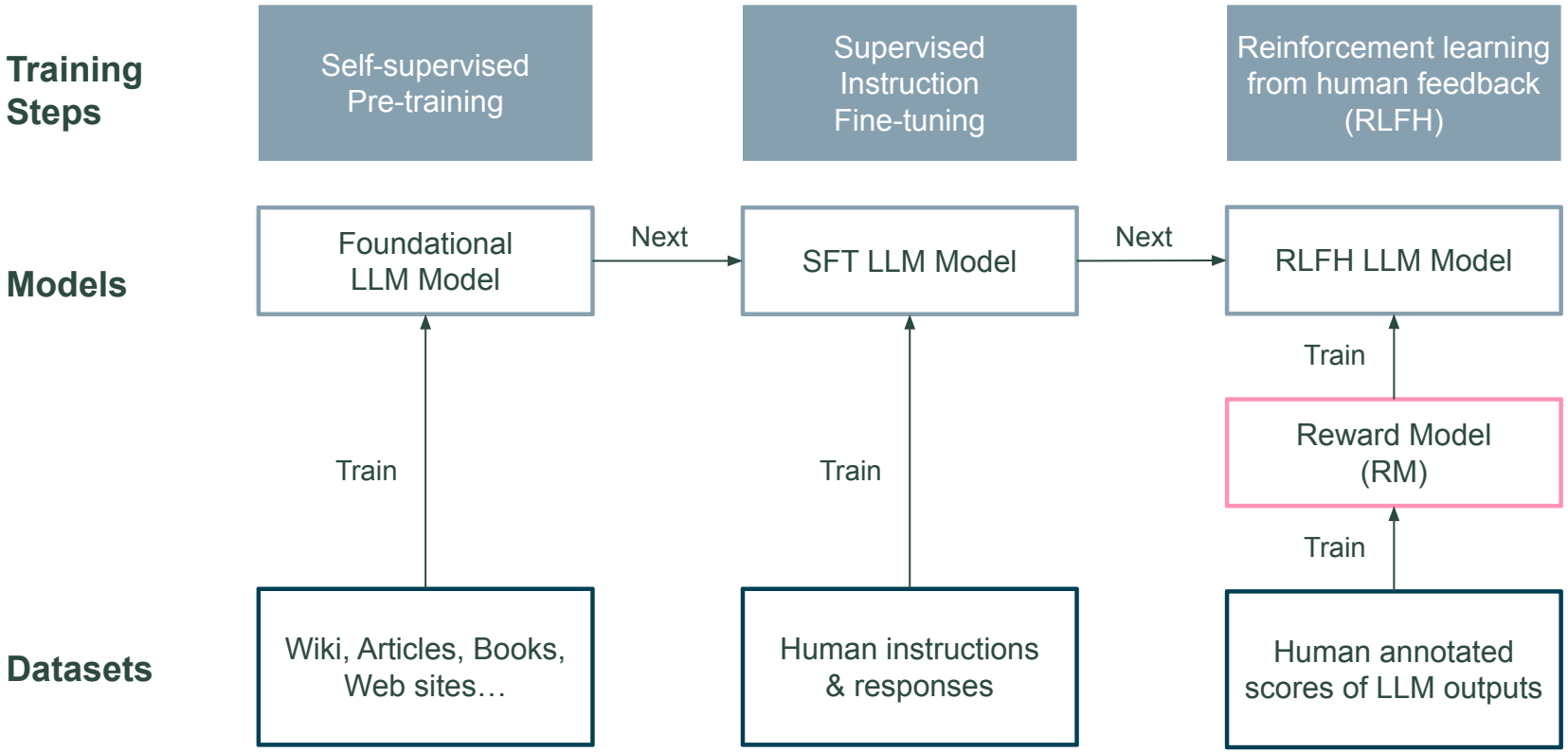
- Model encodes the input sequence
- Computes logits of each possible next token
- **Temperature controls randomness**
    - Low → deterministic, focused
    - High → diverse, creative, less precise
- Logits are transformed to probabilities via softmax



Strategies to choose tokens

- Pick the most likely token
- Sample from top-k candidates
- Beam search - explore multiple possible continuations
- …

How LLMs Are Trained



LLMs: Key Takeaways

LLMs operate on tokens (including special ones), not words or meaning

- Billing and context limits
- Prompt structure and context formatting matter

LLMs predict the next token based on probabilities

- temperature
- hallucinations
    - We automate everything that can be automated.
    - And we AI-ify everything that can't.

Strengths:

- Language tasks (summarization, paraphrasing, text/code generation)
- Structured reasoning patterns, tool use, file/table processing

Limitations:

- Precise algorithmic/symbolic tasks
- Up-to-date knowledge (post-training events)
- Long multi-step planning and persistent memory

➡️ Many limitations can be mitigated in agents using tools and memory.

Школа анализа данных

Thank you!

Zaytseva Alena